

No Man's Land

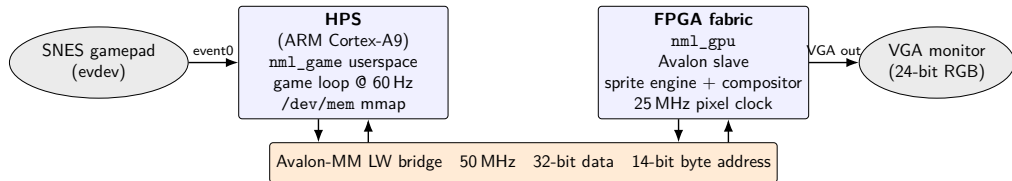
A custom SystemVerilog GPU and game on the DE1-SoC

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CSEE 4840 — Embedded System Design — Spring 2026

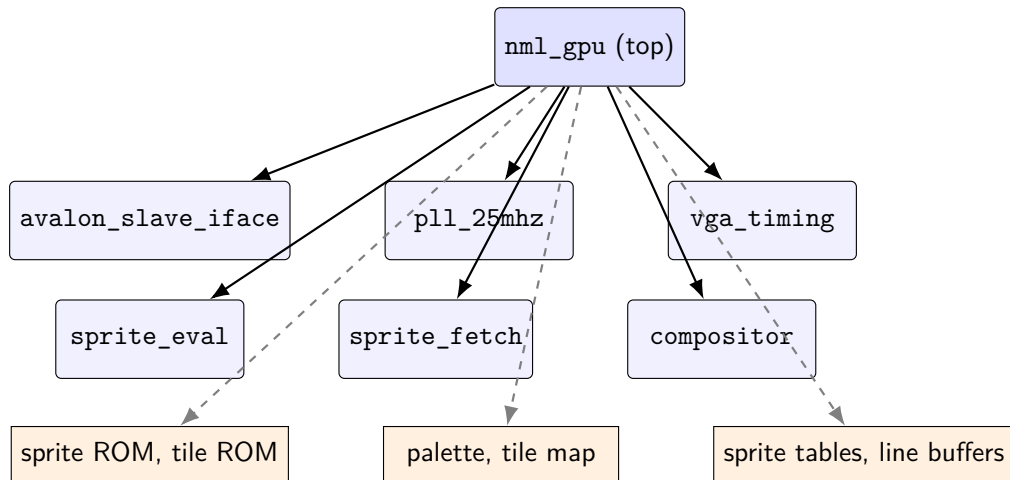
Instructor: Prof. Stephen Edwards

System block



HPS writes shadow state; FPGA latches at VSYNC; pixels always stream.

What's inside nml_gpu

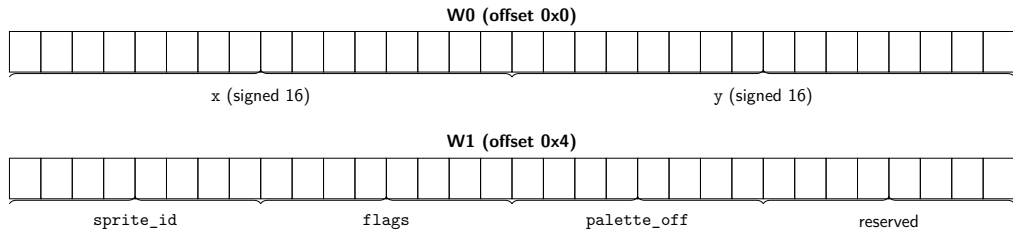


One 50 MHz domain (Avalon) + one 25 MHz domain (pixel). All RAMs/ROMs inferred as M10K.

Avalon register map — 16 KB window at 0xFF200000

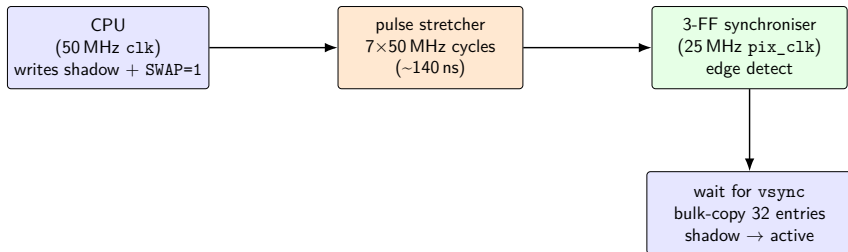
Offset	Name	Layout
0x0000	CTRL	[0]=ENABLE, [1]=SWAP (auto-clear), [2]=HUD_ON
0x0004	STATUS	[0]=VBLANK, [1]=SWAP_PENDING
0x0008	BG_SCROLL	[15:0]=scroll_x, [31:16]=scroll_y (signed)
0x000C	IRQ_MASK	[0]=VBLANK IRQ enable
0x0010	PLAYER_POS	[15:0]=px, [31:16]=py
0x0014	PLAYER_STATS	[7:0]=hp, [15:8]=wave (BCD), [31:16]=level
0x0018	SCORE	[23:0]=6-digit BCD score
0x001C	KILL_COUNT	32-bit kills
0x0020	HUD_AUX	[7:0]=ammo BCD, [11:8]=art, [15:12]=gas
0x0100-0x01FF	Sprite table	32 entries × 8 B
0x0400-0x07FF	Palette	256 entries × 4 B (RGB888)
0x1000-0x22BF	Tile map	80 cols × 60 rows, 1 byte/tile

Sprite-table entry — 8 bytes / 64 bits



flags byte: [2] hflip [3] vflip [6:4] priority (0 = highest) [7] ACTIVE

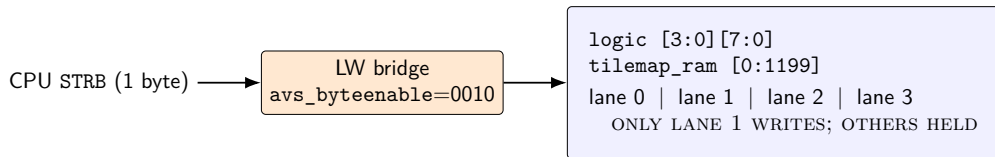
Frame commit — shadow → active at VSYNC



CPU then polls `STATUS.SWAP_PENDING`;
clears one cycle after the bulk copy.

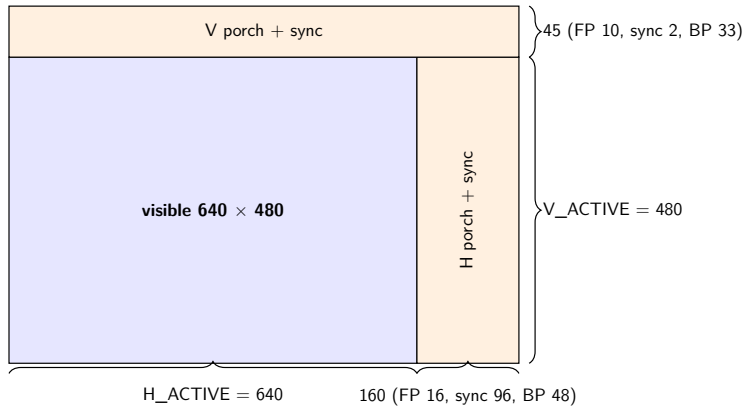
One 20 ns Avalon pulse would slip past the 40 ns pixel clock. Widen to 140 ns and the CDC is deterministic.

Tile-map byte writes — packed [3:0] [7:0]



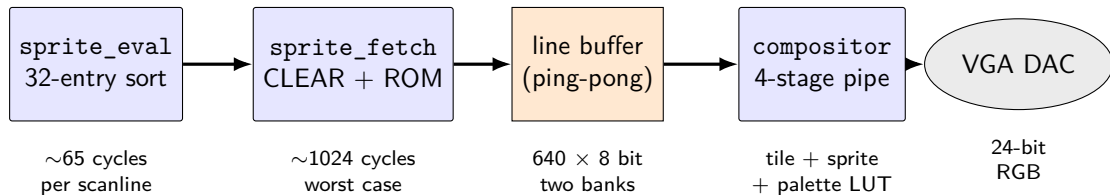
Why packed? Byte-organised storage forced bridge RMW to clobber 3 lanes per word. Packed inference uses M10K `byteena_a`: one lane updates, the other three are untouched.

VGA timing — 640×480 @ 60 Hz, 25 MHz dot



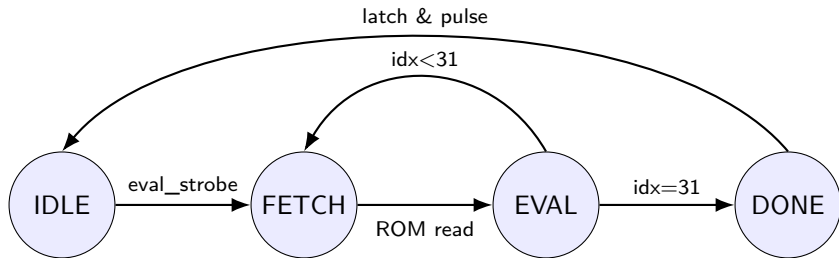
One frame = 800×525 dot cycles. HBLANK is the 160-cycle window we use to evaluate sprites for the next scanline.

Pixel pipeline — sprite engine + compositor



Eval runs inside HBLANK (160 cycles). Fetch spans HBLANK + visible (800 cycles). Line buffers are double-buffered, so fetch lags read by one scanline. Compositor never stalls.

sprite_eval — per-scanline priority sort



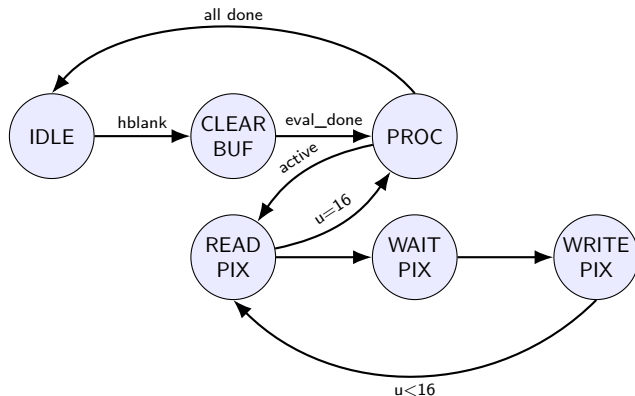
Per sprite (1 cycle in EVAL):

`intersects = active && y ≤ next_scanline < y+16`

insertion-sort into 8 priority-ordered slots (all 8 compares in parallel)

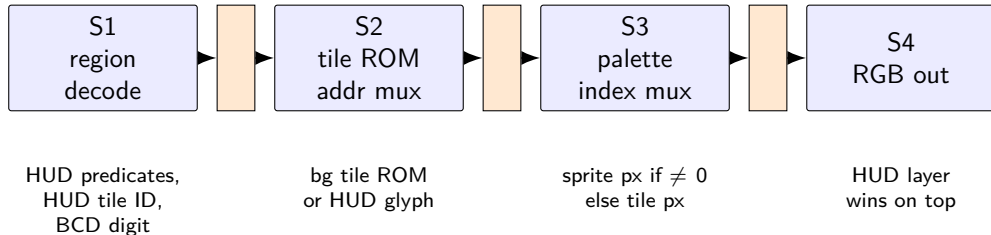
Total: 32 fetch+eval cycles + 1 DONE = ~65 cycles. Fits easily in 160-cycle HBLANK.

sprite_fetch — ROM read + line-buffer fill



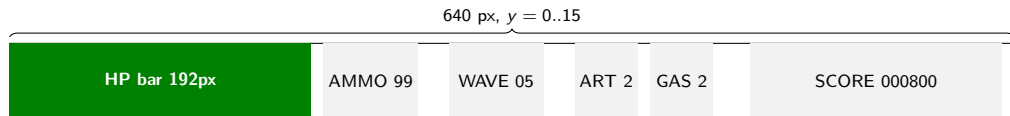
Worst-case cycle budget per scanline: $\text{CLEAR_BUF (640)} + 8 \text{ sprites} \times 48 \text{ (384)} \approx 1024$ cycles.
HBLANK alone is 160; HBLANK + visible is **800**. Fetch lives one line behind the read, so the ping-pong line buffers absorb the latency.

Compositor — 4-stage pipeline



Latches between stages absorb the 1-cycle latency of tile ROM and palette M10K reads. Never stalls.

HUD strip — top 16 pixels of every frame



HP bar: 2 px per HP, clamped to 192. Glyphs: 8×8 from tile ROM, slots 16–41 (A–Z), 48–57 (0–9). Scores, ammo, wave packed BCD in registers.

Memory inventory (all inferred M10K)

Memory	Shape	Init	Bits
sprite ROM	$16,384 \times 8$	\$readmemh	131,072
tile ROM	$4,096 \times 8$	\$readmemh	32,768
tile map	$1,200 \times 32$ (packed 4×8)	blank (HPS writes)	38,400
palette	256×24	\$readmemh + HPS overrides	6,144
sprite table act.	32×64	via VSYNC bulk copy	2,048
sprite table shd.	32×64	HPS writes	2,048
line buffer A	640×8	cleared per HBLANK	5,120
line buffer B	640×8	cleared per HBLANK	5,120
Total			86,144 bits

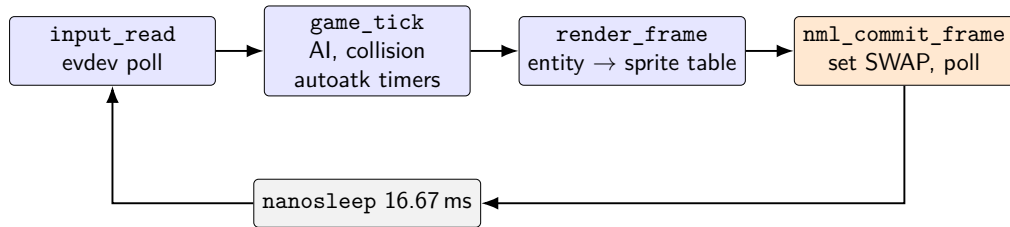
14 M10K blocks out of 397.

Resource utilisation — Cyclone V 5CSEMA5F31C6

Resource	Used	Available	%
ALMs	205	32,070	< 1%
Registers	365	—	—
Pins	54	457	12%
Block memory bits	86,144	4,065,280	2%
M10K blocks	14	397	4%
DSP blocks	0	87	0%

Compile time: 63 s wall clock (Analysis 13 s, Fitter 30 s, Asm 12 s, STA 8 s).

Software architecture — 60 Hz game loop on the ARM



game_t holds a 64-entity pool. State machine: PLAYING ↔ LEVELUP → GAMEOVER.
Mortar is timer-driven; artillery + gas fire from L/R shoulders.

What actually works

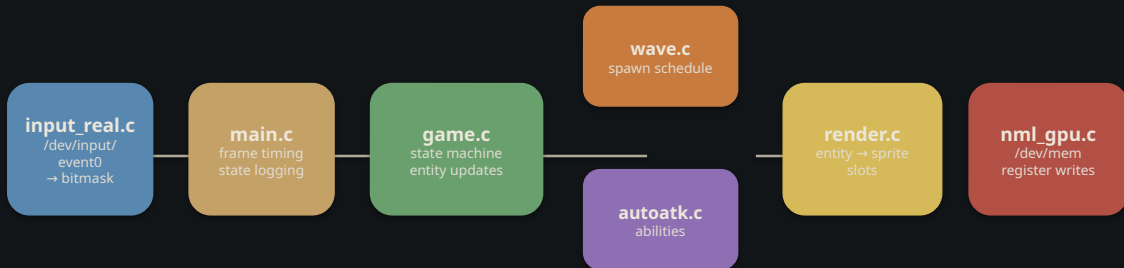
640×480 @ 60 Hz VGA on the DE1-SoC monitor
Tile-mapped battlefield (80×60 from 64-tile ROM)
32 hardware sprites, top-8 sorted per scanline, hflip
256-entry RGB888 palette, M10K-inferred
HUD strip: HP bar, AMMO, WAVE, ART, GAS, SCORE (BCD)
DragonRise SNES gamepad over evdev (/dev/input/event0)
60 Hz HPS game loop driving sprite-table writes
Frame-commit handshake (CTRL.SWAP → VSYNC bulk copy)
Mortar timer, artillery column kill + beam, mustard gas cloud
Wave spawner, level-up menu, ammo drops, game-over screen

No Man's Land

Program Deep Dive: from controller input to VGA sprite table

What the software actually does each frame

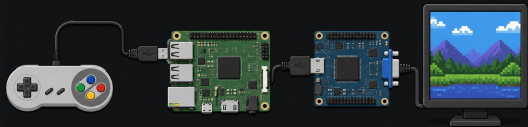
File-level responsibilities with the real call path



The main separation is intentional: gameplay logic never knows whether output is terminal text or the real FPGA.

Two builds, one game core

The same `game_t` and `game_tick()` are reused



Terminal build

`render_terminal.c`
prints sprite slots

FPGA build

`render.c` + `nml_gpu.c`
writes registers

Shared core

`main.c` → `input_read()` → `game_tick()` → `render_frame()`

```
uint16_t input = input_read(game.frame);
game_tick(&game, input);
render_frame(&game);
```

The game state is just structured memory

game_t owns every object and counter

game_t

ents[64]
player_i
frame
score / hp
state / prev_state
prev_input
wave counters
autoatks[]
ammo / charges
beam visual state

entity_t

kind
active
x, y
vx, vy
hp
phase
ttl / ttl_max
payload
fire_cd

Entity kinds

PLAYER
ENEMY_ARMED
ENEMY_UNARMED
BULLET
ENEMY_BULLET
AUTO_PROJ
HAZARD
AMMO_DROP

main.c controls timing, not game rules

The loop handles hardware init, signals, frame duration, and debug prints

```
if (!NML_TERMINAL_BUILD) {  
    nml_open();  
    init_palette_runtime();  
    render_init_tilemap();  
    nml_set_enable(1);  
}  
  
while (g_running) {  
    input = input_read(frame);  
    game_tick(&game, input);  
    render_frame(&game);  
}
```

startup

FPGA path maps /dev/mem and paints palette/tilemap

per-frame

input → tick → render

transitions

prints wave clear, level-up, game-over logs

shutdown

SIGINT/SIGTERM turns off video cleanly

input_real.c turns Linux events into a bitmask

No game logic depends on Linux input details



Control	Linux code	Game bit
D-pad X/Y	ABS_X / ABS_Y	LEFT / RIGHT / UP / DOWN
B / Y / X / A	BTN_THUMB2 / TOP / TRIGGER / THUMB	fire down / left / up / right
L / R	BTN_TOP2 / BTN_PINKIE	artillery / gas
Start	BTN_BASE4	restart / confirm path

game_tick() is a state machine

Most bugs are easier to locate once the state transition is clear

STATE_PLAYING

updates entities
checks wave clear

STATE_LEVELUP

cursor move
B confirms

STATE_GAMEOVER

wait for Start
then game_init()

Every tick starts by saving prev_state and increments frame; prev_input is stored at the end for edge detection.

wave_advance()

applies upgrade then resets wave counters

Player update: movement, clamps, firing, abilities

update_player() consumes the bitmask produced by input_real.c

Move

D-pad adds ± 4 px/frame
to x/y

Clamp

x: 0..624
y: HUD_H..464

Fire

face buttons choose
axis-aligned bullet

Abilities

L: artillery
R: gas cloud

```
if (INPUT_FIRE_DOWN) bullet( 0, +8);  
else if (INPUT_FIRE_LEFT) bullet(-8, 0);  
else if (INPUT_FIRE_UP) bullet( 0, -8);  
else if (INPUT_FIRE_RIGHT) bullet(+8, 0);
```

```
if (fire_cooldown != 0) return;  
if (ammo <= 0) return;  
spawn ENT_BULLET;  
fire_cooldown = 5;  
ammo--;
```

Enemy AI is simple but intentionally de-synchronized

Armed enemies add fire behavior on top of movement

Vertical pressure

e->y steps toward player row
at wave speed

Horizontal choice

25% left, 25% right,
50% chase player

Armed firing

cooldown counts down
then aims once

```
r = enemy_rng(frame, phase)
if r < 64: x -= speed
elif r < 128: x += speed
else: x → player.x
```

```
dx = player.x - enemy.x
dy = player.y - enemy.y
norm = |dx| + |dy|
v = 3 * (dx,dy) / norm
```

```
phase = rand() & 0xff
fire_cd = 60 + rand()%120
// avoids synchronized waves
```

Motion updates are separate from collision handling

Each entity type has one simple update rule

Player bullet

$x += vx$, $y += vy$
despawn off-screen

Enemy bullet

same motion path
damages player later

Ammo drop

$t_{tl}--$ until 0
collected by AABB

Beam

$beam_ttl$ controls
tile column

This separation keeps behavior predictable: movement first, then a single collision pass resolves consequences.

```
update_enemies(); update_bullets(); update_enemy_bullets(); update_ammo_drops(); autoatk_tick(); autoatk_update_proj(); handle_collisions
```

No collision is applied during the motion loops

Collision handling is ordered by gameplay meaning

handle_collisions() owns damage, pickups, and score bookkeeping

1	Ammo drop ↔ player	ammo += 10, cap at 99, drop disappears
2	Enemy bullet ↔ player	player HP -10, bullet disappears
3	Enemy ↔ player	player HP -10, enemy disappears, no score
4	Player bullet ↔ enemy	bullet disappears, apply_damage(enemy, 1)
5	Auto hazard/projectile ↔ enemy	damage + optional despawn policy by weapon kind

All enemy kills route through apply_damage() so score, kill counters, and ammo drops stay consistent.

wave.c is a data-driven spawner

Difficulty lives in the WAVES table

Wave	period	total	armed	armed HP	speed
1	60	6	0	1	1
2	50	8	1	2	1
3	45	10	2	2	2
4	35	12	4	2	2
5+	25	16	6	3	3

wave_tick()

spawn when cooldown hits zero

alloc_entity()

find inactive entity slot

initialize enemy

kind, hp, phase, fire_cd

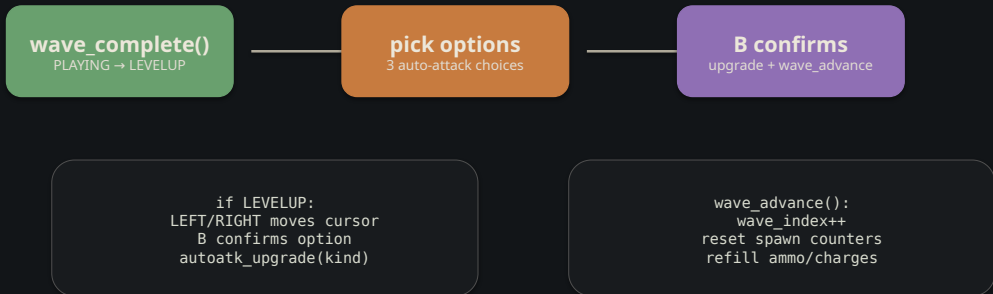
wave_complete()

spawned all + none alive

Wave 5 clamps and repeats as the hardest pattern

Level-up pauses the simulation before advancing the wave

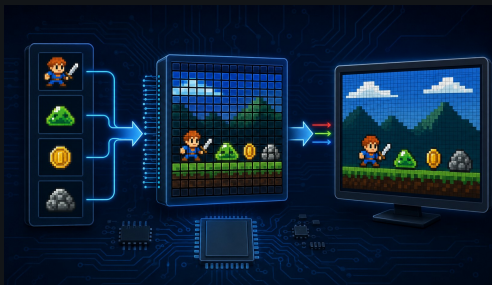
The upgrade menu is still part of game_tick()



Important detail: `wave_advance()` happens after the player picks an upgrade, not immediately when the wave ends.

render.c converts entities into hardware sprite slots

This is the bridge from gameplay state to VGA-visible objects



slot 0

always player

slots 1-31

entities in array order

gas hazard

expands into 1-4 sprites

leftover slots

`nml_hide_sprite()`

```
emit_sprite(&slot, e->x, e->y,  
            entity_to_sprite_id(e),  
            entity_palette_off(e),  
            priority);
```

nml_gpu.c packs C structures into Avalon register writes

The program writes a 16 KB LW bridge window mapped at 0xFF200000

Region	Offset	Write meaning
CTRL	0x0000	ENABLE / SWAP / HUD_ON
PLAYER_POS	0x0010	x and y packed into one word
PLAYER_STATS	0x0014	hp + BCD wave + level
SCORE	0x0018	HUD score
HUD_AUX	0x0020	ammo + artillery + gas charges
SPRITE_TABLE	0x0100	32 × 8-byte sprite entries
PALETTE	0x0400	RGB888 entries
TILEMAP	0x1000	80 × 60 tile IDs

nml_open()

`open("/dev/mem") → mmap(NML_LWFPGA_BASE)`

nml_write_sprite()

`W0 = y:x, W1 = sprite_id:flags:palette_off`

nml_commit_frame()

sets CTRL.SWAP so hardware commits at VSYNC

One sprite slot is two 32-bit words

This is what render.c ultimately writes per visible object

W0: position

bits 15:0 = x bits 31:16 = y

W1: identity + draw flags

sprite_id, flip bits, priority, active, palette offset

Field	Purpose
sprite_id	selects 16×16 art in sprite ROM
ACTIVE	visible / hidden
priority	scanline culling order
hflip / vflip	future sprite mirroring
palette_off	recoloring hook

```
w0 = ((uint32_t)(uint16_t)y << 16) | (uint16_t)x;  
w1 = sprite_id | (flags << 8) | (palette_off << 16);
```

render_terminal.c is a diagnostic mirror of render.c

It helps verify slot allocation before touching the FPGA

mock_sprite_t

active, x, y, sprite_id, glyph

entity_glyph()

P, A, u, *, ^, +, M, ~

slot printout

frame, hp, score, wave, ammo, 32 slots

```
frame=150 hp=100 score=0 sprite_count=3
slot 00: sprite_id=1 x=320 y=420 glyph=P
slot 01: sprite_id=2 x=320 y=115 glyph=A
slot 02: sprite_id=4 x=53 y=55 glyph=u
        slot 03: inactive
```

Where to debug when something goes wrong

Symptoms map to specific files

Symptom	First place to look
controller moves nothing	input_real.c: event device, mapping, permissions
player moves but cannot fire	update_player(): ammo + fire_cooldown
enemies sync into columns	phase / enemy_rng() in update_enemies()
wave never ends	wave_complete(): spawned count + active enemies
sprites ghost on screen	render.c: hide leftover slots
FPGA shows blank	nml_open(), CTRL.ENABLE, palette/tilemap init
HUD wrong	nml_set_player_state(), nml_set_score(), HUD_AUX